

Azure Data Processing Stages

Generation & Ingestion

Generation

Generation: Data is created across different source systems, such as IoT devices, web and mobile apps, or enterprise transactional databases, OLTP.

Ingestion

Ingestion: The process of capturing and importing data from various sources.

Batch Ingestion

Batch Ingestion: Collects and moves data in large, scheduled blocks (e.g., end-of-day sales).

Streaming Ingestion

Streaming Ingestion: Captures data continuously and in real time as events occur (e.g., financial fraud detection or live GPS tracking).

Storage

- The location where ingested data is safely kept and organized for future access.

Data Lakes

- Store vast amounts of raw, unstructured, and structured data in its native format.

Data Warehouses

- Highly structured, optimized repositories designed specifically for fast SQL queries and business intelligence.

Processing & Transformation

- The engine that cleans, organizes, and transforms raw data into actionable insights and accomplished through either an ETL or ELT pipeline.

Batch Processing

- Processes large, previously collected datasets periodically.

Stream Processing

- Analyzes real-time data on-the-fly with near-zero latency.

Modeling

- Involves designing structural schemas—such as star or snowflake schemas—that organize the transformed data. This step ensures data scalability and faster query performance for reporting and BI tools.

Delivery & Consumption

- The final layer where end-users and applications access the processed data. This includes dashboards (like PowerBI or Tableau), machine learning models, and automated APIs.
- The structured, high-quality data is made available to business analysts, data scientists, and AI/ML models for decision-making.

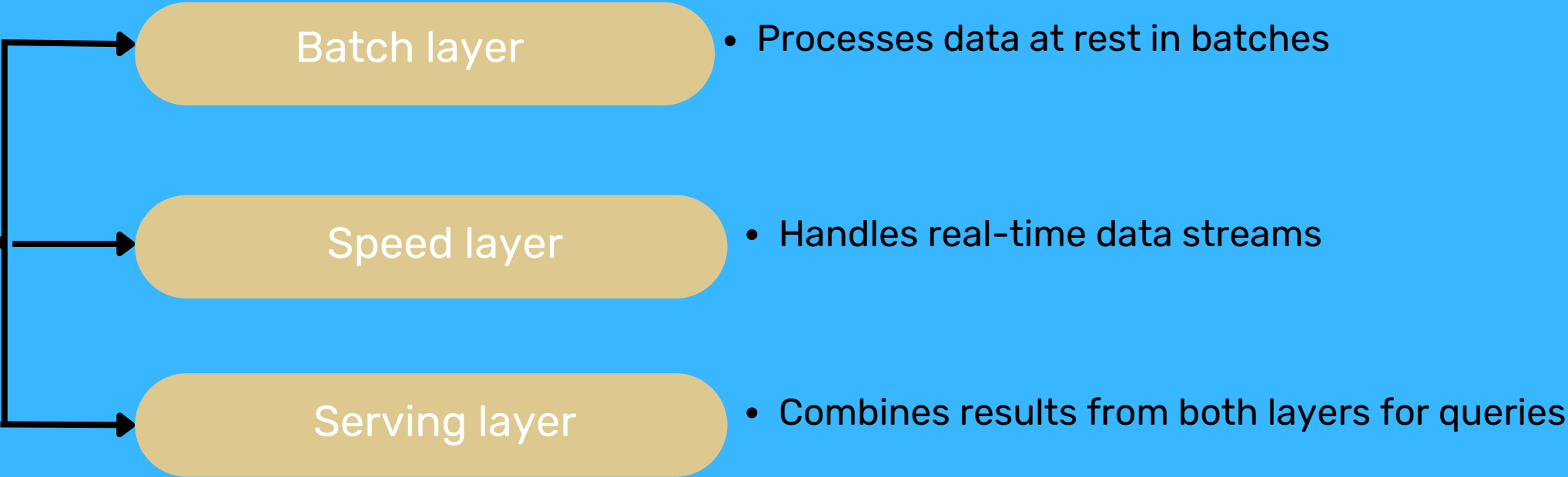
Ongoing Monitoring & Maintenance

- A continuous process that ensures the pipeline remains reliable, cost-effective, and compliant with data privacy regulations.
- This includes orchestrating workflows, maintaining data lineage, and conducting data quality testing to catch pipeline issues early.

Azure Data Processing Architectural Design Patterns

Lambda

- A traditional hybrid approach that combines batch processing and stream processing to handle massive data volumes with low latency. It has three layers.



Kappa

The Kappa Architecture is a modern big data processing pattern that simplifies the data architecture by removing the batch layer completely, treating all data processing as an ongoing stream. Best for:

- IoT data processing
- Systems where reprocessing historical data is rare.

Medallion

- A medallion architecture is a data design pattern used to organize data logically in a Lakehouse. Its goal is to incrementally and progressively improve the structure and quality of data through layers:



Batch Processing

- Processes data in fixed groups at scheduled intervals (e.g., daily/weekly batches)

Stream Processing

- Processes data continuously in real-time as it arrives

Data Warehouse

- Structured data for BI/reporting

Data Lake

- Raw data storage in native format

Data Lakehouse

- Combines data lake flexibility with warehouse management

Data Mesh

- Decentralized, domain-oriented data ownership

Event-Driven

- Real-time fraud detection, monitoring

- Lambda and Kappa are the most specific data-processing architectures for handling big data with varying latency requirements.